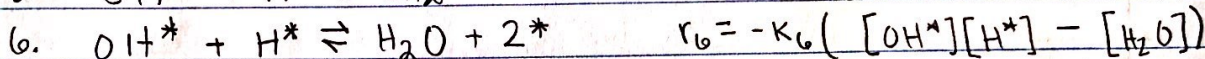
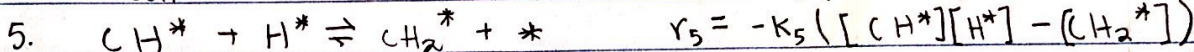
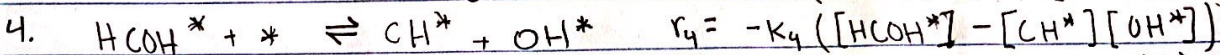
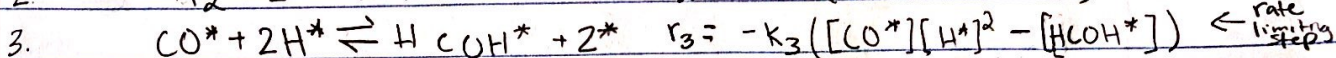
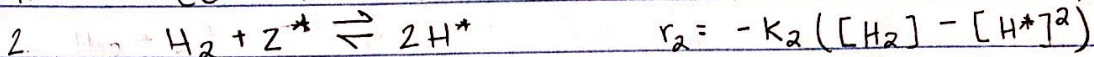
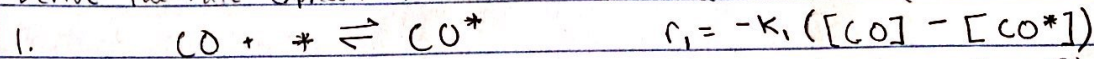


1. Derive the rate expression for FT-VIII, if the third rxn (in FT-VIII) is the rate limiting step.



$$\frac{r_1}{k_1} = \frac{r_2}{k_2} = \frac{r_4}{k_4} = \frac{r_5}{k_5} = \frac{r_6}{k_6} = 0 \Rightarrow \textcircled{1} [\text{CO}] = [\text{CO}^*]$$

$$\textcircled{2} [\text{H}_2] = [\text{H}^*]^2$$

$$[\text{HCOH}^*] = [\text{CH}^*][\text{OH}^*]$$

$$\textcircled{4} [\text{HCOH}^*] = [\text{CH}^*][\text{OH}^*]$$

$$\textcircled{5} [\text{CH}_2^*] = [\text{CH}^*][\text{H}^*]$$

$$\textcircled{6} [\text{H}_2\text{O}] = [\text{OH}^*][\text{H}^*]$$

$$r_3 = -k_3([\text{CO}^*][\text{H}^*]^2 - [\text{HCOH}^*])$$

$$= -k_3([\text{CO}][\text{H}_2] - [\text{CH}^*][\text{OH}^*])$$

$$= -k_3([\text{CO}][\text{H}_2] - \frac{[\text{CH}_2^*][\text{H}_2\text{O}]}{[\text{H}^*][\text{H}^*]})$$

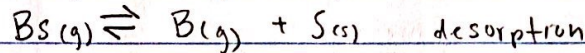
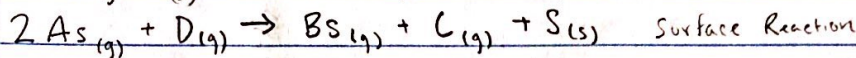
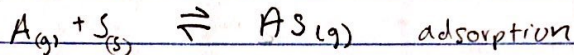
$$r_3 = -k_3 \left([\text{CO}][\text{H}_2] - \frac{[\text{CH}_2^*][\text{H}_2\text{O}]}{[\text{H}_2]} \right)$$

2.



$$-r_A = \frac{K \cdot P_D P_A^2}{(1 + K_A P_A + K_B P_B)^2}$$

Proposed mechanism:



surface Reaction is the rate limiting:

$$r = K (C_{AS}^2 P_D)$$

For adsorption and desorption we have

$$C_{AS} = K_A P_A C_V$$

$$C_{BS} = K_B P_B C_V$$

site Balance:

$$C_T = C_V + C_{AS} + C_{BS} \rightarrow$$

$$C_V = \frac{C_T}{1 + K_A P_A + K_B P_B}$$

$$\rightarrow r = \frac{K P_D P_A^2}{(1 + K_A P_A + K_B P_B)^2}$$

$$\text{rate of adsorption} = -K_{ad} (P_A C_V - \frac{1}{K_A} C_{AS})$$

$$\text{rate of desorption} = -K_d (C_{BS} - K_B P_B C_V)$$

$$\frac{r_{ad}}{K_{ad}} = \frac{r_{des}}{K_d} = 0$$

$$\begin{aligned} C_{AS} &= K_A P_A C_V \\ C_{BS} &= K_B P_B C_V \end{aligned}$$

$$-r = K \left(\frac{K_A P_A C_T}{1 + K_A P_A + K_B P_B} \right)^2 P_D \xrightarrow{\text{Simplify}} \frac{K \cdot P_D P_A^2}{(1 + K_A P_A + K_B P_B)^2}$$

$$-r_A = \frac{K \cdot P_D P_A^2}{(1 + K_A P_A + K_B P_B)^2} \quad \checkmark$$